

HARIPRASAD RAVISHANKAR

PRINCIPAL ENGINEER & SENIOR TEAM LEAD

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WORK EXPERIENCE

MathWorks, Principal Engineer & Senior Team Lead

May 2026 - Present

- Guide a compiler team towards advanced deep learning optimizations (MLIR/CGIR pipeline).
- Launched the **MATLAB Coder Support Package for PyTorch and LiteRT Models (R2026a)**.
- Maintainer of **Predict**, **Stateful Predict**, **Stateful Classify**, and **Classify** blocks in the Deep Learning Toolbox's Simulink library, supporting robust deployment pipelines.

MathWorks, Senior Team Lead

May 2023 - May 2026

- Led a team of 3 software engineers managing DL compiler roadmap and technical design.
- Directed code generation solutions targeting Intel, ARM, and NVIDIA based hardware architectures.
- Defined product strategy for DL compilers; presented bi-annually to Executive VPs.

MathWorks, Senior Software Engineer

May 2021 - May 2023

- Architected C++/CUDA code generation from PyTorch, TFLite, and MATLAB frameworks.
- Drove deep learning code-gen integration in Simulink for Model Reference and Rapid Accel.

MathWorks, Software Engineer - Deep Learning Compilers

Mar 2019 - May 2021

- Wrote efficient, optimized implementations of deep learning layers and compiler operations.
- Designed graph-level compiler optimizations (e.g., operator fusions) to accelerate inference.
- Deployed DL models on target hardware including NVIDIA Jetsons, Raspberry Pi, ARM Mali, and Intel.

MathWorks, Software Engineer in Test

May 2018 - Mar 2019

- Enhanced testability and wrote validation tests for GPU Coder and DL code generation on FPGAs, GPUs, and ARM.

MathWorks, Application Support Engineer

May 2017 - Apr 2018

- Technical specialist resolving neural network, ML, and optimization issues for commercial and academic customers.

NinePoint Medical, Deep Learning Co-op

Jan 2017 - May 2017

- Developed U-Net based deep learning architectures for segmenting medical feature structures in OCT scans.
- Implemented verification pipeline and post-processing; features successfully received **FDA clearance**.

Philips Research, Research Intern

Jun 2016 - Jul 2016

- Created AlexNet and deconvolutional segmentation models for cervical cancer nucleus detection.
- Achieved a PPV of 91% and Jaccard Score of 95% on unseen data validation sets.

EDUCATION

MS, Electrical Engineering

University of Southern California
GPA 3.86/4.0 | 2015 - 2017

B.Tech, Electronics

Vellore Institute of Technology
CGPA 9.13/10 | 2011 - 2015

SKILLS

Programming: C++, MATLAB, Simulink, CUDA, Python, PyTorch, LiteRT/TFLite, MLIR, CGIR

Libraries & Tools: TensorRT, cuDNN, cuBLAS, OpenCV, Linux, Git

Leadership: Technical Strategy, Roadmap Planning, Team Management

INITIATIVES

Compiler IR Framework (MLIR/CGIR)

Lowering pipeline streamlining transition from PyTorch to optimized CUDA.

AI Interoperability Product

Direct C++/CUDA code-gen support for PyTorch/LiteRT in MATLAB R2026a.

Hardware-Aware SIMD Optimizations

Vectorized kernels utilizing loop scheduling and cache-blocking for ARM & x86.

Model Compression & Specialized Types

Implemented network pruning (taylorPrunableNetwork) and BF16 solutions.

High-Level AI Primitive Design

End-to-end dlarray code generation with dynamic memory handling.